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Editorial Office

Dear Editor,

I submit the manuscript “**A Worked Identifiability Audit of a Stipulated Preparation-Indexed Potential in a Homogeneous Bose–Einstein Condensate**” for consideration as a methods or tutorial note.

The paper examines a deliberately underdetermined construction: a static condensate potential is stipulated to depend on a preparation’s quantum relative entropy. It separates the exact Gibbs identity, the free state-to-potential map, the fixed-particle-number stationary response, and the statistical inference contract. The manuscript does not present their composition as a new physical interaction.

The numerical revision makes the particle-number convention explicit and validates the same localized profile against independent imaginary-time Gross–Pitaevskii relaxation. A deterministic two-bin fixture demonstrates both an identifiable nuisance model and complete template absorption. A phase-randomized classical-field example is retained only as generic motivation for controls; it is not presented as a matched thermal preparation, covariance estimate, or uncertainty floor.

The central result is cautionary: a preparation contrast cannot identify the free coefficient until seven declared contracts cover the ensemble, intervention, template and conservation convention, preparation matching, nuisance basis, measurement covariance, and decision rule. The note is intended for advanced students and researchers learning to distinguish algebraic correctness, reproducibility, identifiability, and physical evidence. It makes no analogue-gravity, detection, or gravitational claim.

Declarations. This manuscript has not been published elsewhere and is not under consideration by another journal. The author has no competing interests to declare. AI tools from Anthropic and OpenAI were used for copy editing, code review, and literature synthesis; the author reviewed and takes responsibility for all scientific claims, calculations, citations, and conclusions.

Thank you for considering this manuscript.

Sincerely,

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